QUADRATIC PERSISTENCE OF SEGRE VARIETIES AND A LINEAR GAP BETWEEN BIQUADRATIC SOS RANK AND ZARANKIEWICZ BOUNDS

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Abstract. We compute the quadratic persistence of the Segre varieties $\mathbb{P}^{m-1} \times \mathbb{P}^2$ and $\mathbb{P}^{m-1} \times \mathbb{P}^3$ for every $m \geq 2$: the values are m and $\lceil 4m/3 \rceil$. By the Pythagoras-number bound of Blekherman, Sinn, Smith and Velasco, some $m \times 4$ biquadratic sums of squares require at least $\lfloor 8m/3 \rfloor$ real bilinear squares. This exceeds both the second order and the augmented second order Zarankiewicz numbers at $m = 21$ and for every $m \geq 23$, and the gap is at least $m/6 - O(1)$. In every width $n \geq 3$ we prove $\mathrm{qp} \leq n\lceil m/3 \rceil$, so the gap is at least $\frac{n-3}{6}m - O_n(1)$ for fixed $n \geq 4$, and we certify exact values in widths five and six, where they meet the lower bound of the same authors. We also construct rational forms attaining the persistence lower bound whose rational length exceeds their real length by exactly three. A specialization of the parallel-line covering bound of the same authors gives $\mathrm{BSR}(m,n) \leq mn - \max(m,n) + 1$, hence $2m \leq \mathrm{BSR}(m,3) \leq 2m+1$ and $8 \leq \mathrm{BSR}(4,3) \leq 9$.

1. Introduction

A biquadratic form $f(x,y)$, $x \in \mathbb{R}^m$, $y \in \mathbb{R}^n$, is a sum of squares if $f = \sum_{k \leq r} b_k^2$ with b_k bilinear; let $\ell_K(f)$ be the least such r with b_k over a field K. For $m, n \geq 2$ the invariant $\mathrm{BSR}(m,n)$ is the maximum, over real sum-of-squares biquadratic forms of size $m \times n$, of $\ell_{\mathbb{R}}(f)$ [3]. It is studied through graph parameters: Cui, Qi and Xu showed $\mathrm{BSR} \geq z$ for the Zarankiewicz number z [3], Xu, Cui and Qi found $\mathrm{BSR}(4,3) \geq 8 > z(4,3)$ [8], and Löfberg and Qi introduced the second order Zarankiewicz number z_2 with $\mathrm{BSR} \geq z_2$ [5, Theorem 3.4]. In width four Chen and Chen proved $z_2(m,4) = \lfloor (5m+6)/2 \rfloor$ for $m \geq 15$ and noted that this does not determine $\mathrm{BSR}(m,4)$ [2, Theorem 1.1 and §5.2]. Qi and Cui showed $\mathrm{BSR}(4,4) \geq z_{2A}(4,4) = 11 > z_2(4,4)$ [7], so BSR and z_2 already differ there; what is new here is a gap that grows linearly.

Bilinear forms are the linear forms on the Segre variety $X \subset \mathbb{P}^{mn-1}$, so $\mathrm{BSR}(m,n)$ is the Pythagoras number of X, and since X is non-degenerate, irreducible and totally real, [1, Theorem 1.3] gives $\mathrm{BSR}(m,n) \geq mn - \mathrm{qp}(X)$. We compute $\mathrm{qp}(X)$ for $n = 3, 4$. In width four the resulting lower bound has slope $8/3$, while every graph parameter above is at most $5m/2 + O(1)$ by the cell bound, so $\mathrm{BSR}(m,4) - z_{2A}(m,4) \geq m/6 - O(1)$. The exact slope of BSR itself remains open. We have not found these values of qp or this comparison in the literature; our search covered the papers cited here and is not a proof of priority.

Date: November 2026.

2. Quadratic persistence of a Segre variety

Let $A=\mathbb{R}^m$, $B=\mathbb{R}^n$ and $X=\mathbb{P}(A)\times\mathbb{P}(B)$ in $\mathbb{P}(A\otimes B)$. Let $\mathcal{D}$ be the space of symmetric bilinear forms D on $A\otimes B$ with $D(z,z)=0$ for every simple tensor z. It is spanned by the 2×2 minors, and identifies with $\bigwedge^2 A^*\otimes\bigwedge^2 B^*$: we write $D(a,a';b,b')$, alternating in (a,a') and in (b,b'), and $d=\dim\mathcal{D}=\binom{m}{2}\binom{n}{2}$. A point $a\otimes b$ lies in the radical of D if and only if $D(a,x;b,y)=0$ for all $x\in A$, $y\in B$. By [1, Definition 3.1 and Lemma 3.3(i),(iv)], $\operatorname{qp}(X)$ is the least k such that some k points of X lie in the radical of no nonzero $D\in\mathcal{D}$, and general points realize it. We use the bound [1, Corollary 3.7]

$$d\le g(s):=sc-\binom{s}{2},\qquad s=\operatorname{qp}(X),\ c=(m-1)(n-1). \tag{1}$$

Since $g(s+1)-g(s)=c-s$, the function g increases strictly for $s<c$. The evaluation matrices are defined over $\mathbb{Q}$ and rational points of X are Zariski dense, so the criterion by minors computes the same persistence over $\mathbb{Q}$, $\mathbb{R}$ or $\mathbb{C}$.

For $D\in\mathcal{D}$ and a basis $e_1,\dots,e_m$ of A put $D_{ij}=D(e_i,e_j;\cdot,\cdot)\in\bigwedge^2 B^*$, so $D_{ji}=-D_{ij}$ and D is determined by the D_{ij}, $i<j$. The point $e_i\otimes b$ lies in the radical of D if and only if $D_{ij}(b,\cdot)=0$ for all j.

Theorem 1. $\operatorname{qp}(\mathbb{P}^{m-1}\times\mathbb{P}^2)=m$ *for* $m\ge 2$.

Proof. Take $b_i=(1,i,i^2)$ and the points $e_i\otimes b_i$. If they lie in the radical of D, then D_{ij} has b_i and b_j in its radical. These are independent, and a nonzero alternating form on $\mathbb{R}^3$ has a radical of dimension one, so $D_{ij}=0$ and $D=0$. Thus $\operatorname{qp}\le m$. Here $c=2(m-1)$ and $d=3\binom{m}{2}$, and $d-g(m-1)=m-1>0$, so (1) excludes $s\le m-1$. This lower bound was known: $\operatorname{qp}(X)\ge\ell(X)$ [1, Theorem 1.2], and the linear strand of a Segre product is nonzero up to $p=m+n-3$ [6, Theorem 4.1], so $\operatorname{qp}\ge m+n-3$; what is new is the equality. □

Theorem 2. $\operatorname{qp}(\mathbb{P}^{m-1}\times\mathbb{P}^3)=\lceil 4m/3\rceil$ *for* $m\ge 2$.

Proof. Upper bound. Write $B=E\oplus\langle w\rangle$ with $\dim E=3$, split $\{1,\dots,m\}$ into consecutive groups of three, the last possibly shorter, and in a basis of E put $b_i=(1,i,i^3)$. Any three b_i are independent, and any six of the tensors $b_i\otimes b_i$ are independent: their coordinates are the monomials $1,t,t^3$, respectively $1,t,t^2,t^3,t^4,t^6$, at distinct positive nodes, and a nonzero combination of k monomials has at most $k-1$ positive roots by Descartes' rule. Take the $m+\lceil m/3\rceil=\lceil 4m/3\rceil$ integer points

$$e_i\otimes b_i\quad(1\le i\le m),\qquad\Big(\sum_{j\in S}e_j\Big)\otimes w\quad\text{for each group } S.$$

Suppose they lie in the radical of D. As above, D_{ij} has b_i,b_j in its radical, so D_{ij} factors through the plane $B/\langle b_i,b_j\rangle$ and $D_{ij}=c_{ij}\operatorname{vol}(b_i,b_j,\cdot,\cdot)$ for a fixed volume form, with $c_{ij}=c_{ji}$ and $c_{ii}=0$. The point of the group S gives, for every i and y, $\sum_{j\in S}c_{ij}\operatorname{vol}(b_i,b_j,w,y)=0$, that is,

$$\sum_{j\in S}c_{ij}b_j\in\langle b_i\rangle, \tag{2}$$

since b_i and the sum lie in E and $w\notin E$. Inside a group, (2) and the independence of its vectors give $c_{ij}=0$. For two groups $U\ne V$ put $T=\sum_{i\in U,j\in V}c_{ij}\,b_i\otimes b_j$. By (2) for $i\in U$ and $S=V$, $T=\sum_{i\in U}\lambda_i\,b_i\otimes b_i$; by (2) for $j\in V$ and $S=U$, with $c_{ij}=c_{ji}$, $T=\sum_{j\in V}\mu_j\,b_j\otimes b_j$. The at most six tensors $b_k\otimes b_k$, $k\in U\cup V$, are independent, so $T=0$, and since the $b_i\otimes b_j$ with $i\in U$, $j\in V$ are independent, all c_{ij} vanish. Hence $D=0$.

Lower bound. For $m = 2, 3$ the claim is $\operatorname{qp} = 3, 4$, which follows from $\mathbb{P}^1 \times \mathbb{P}^3$, a rational normal scroll, so of minimal degree [1, Theorem 3.8], and Theorem 1 by symmetry. Let $m \geq 4$ and $k = \lceil 4m/3 \rceil - 1$, so $m < k$ and $3k < 4m$. For general points $a_i \otimes b_i$, $i \leq k$, the conditions $Ma_i \in \langle b_i \rangle$ on $M\colon A \to B$ are at most $3k$ linear equations in $4m$ unknowns, so there is $M \neq 0$. If $\operatorname{rank} M = r < 4$, at most $m - r$ of the general a_i lie in $\ker M$, and for the other indices $b_i \in \operatorname{im} M$, which allows at most r of the general b_i; so $k \leq m$, a contradiction. Thus M is onto, and $L(a \otimes b) = Ma \wedge b$ maps $A \otimes B$ onto $\bigwedge^2 B$, as decomposable vectors span it. Fix $\bigwedge^4 B \cong \mathbb{R}$ and put $Q(z, z') = L(z) \wedge L(z')$. It is nonzero, since the Plücker form on $\bigwedge^2 B$ is nondegenerate, it vanishes on simple tensors because $Ma \wedge b$ is decomposable, so $Q \in \mathcal{D}$, and $L(a_i \otimes b_i) = 0$ puts the k points in its radical. So k general points do not suffice. The bound $m+n-3 = m+1$ from [1, 6] is weaker for $m \geq 4$, which is why a direct argument is needed. □

Theorem 3. *For $m, n \geq 3$, $\operatorname{qp}(\mathbb{P}^{m-1} \times \mathbb{P}^{n-1}) \leq \min\{n\lceil m/3 \rceil,\ m\lceil n/3 \rceil\}$.*

Proof. By symmetry it suffices to prove $\operatorname{qp} \leq sn$ for $m = 3s$; for other m, project $\mathbb{R}^{3\lceil m/3 \rceil}$ onto A: if π is onto, $\pi(a) \otimes b$ is in the radical of D exactly when $a \otimes b$ is in the radical of $\pi^* D$, and π^* is injective. Split $A = U_1 \oplus \cdots \oplus U_s$ with $\dim U_l = 3$ and put n points in each block, $a_t^{(l)} \otimes b_t^{(l)}$ with $a_t^{(l)} \in U_l$. Since $\bigwedge^2 A^* = \bigoplus_l \bigwedge^2 U_l^* \oplus \bigoplus_{l<l'} U_l^* \otimes U_{l'}^*$, and the radical conditions of the points of U_l only involve the components containing U_l^*, the map $D \mapsto$(radical conditions) is injective as soon as it is injective on each internal component $\bigwedge^2 U_l^* \otimes \bigwedge^2 B^*$, using the points of U_l, and on each mixed component $U_l^* \otimes U_{l'}^* \otimes \bigwedge^2 B^*$, using the points of U_l and $U_{l'}$. Each of these is a nonempty Zariski open condition once it holds for one choice of points, so it suffices to exhibit one choice per component.

Internal. Take points $a_t \otimes b_t$ with $b_1, \ldots, b_n$ a basis of B and any three a_t independent. In the dual basis write $D = \sum_{t<u} D^{tu} \otimes (b_t^* \wedge b_u^*)$ with $D^{tu} \in \bigwedge^2 U^*$. The point $a_t \otimes b_t$ forces $D^{tu}(a_t, \cdot) = 0$ for all u, so D^{tu} has a_t, a_u in its radical and vanishes, as in Theorem 1.

Mixed. Let U, V be two blocks. In U take $a_i \otimes e_i$, $i \in \mathbb{Z}/n$, for a basis e_i of B and any three a_i independent; in V, over $\mathbb{C}$, take $c_t \otimes f_t$ with $c_t = (1, t, t^2)$ and $f_t = \sum_i t^i e_i$, for the n-th roots of unity t. Write the component as $\sum_{i<j} M^{ij} \otimes (e_i^* \wedge e_j^*)$ with M^{ij} bilinear on $U \times V$ and $M^{ji} = -M^{ij}$. The points of U give $M^{ij}(a_i, \cdot) = M^{ij}(a_j, \cdot) = 0$, so $M^{ij} = (a_i \times a_j) \otimes v_{ij}$, where $a_i \times a_j$ is the covector $x \mapsto \det(a_i, a_j, x)$ and $v_{ij} = v_{ji} \in V^*$. Write $v_{ij}(1, t, t^2) = \sum_{h=0}^{2} c_{ijh} t^h$. The points of V give, for each j and each t, $\sum_{i \neq j} \sum_h c_{ijh} t^{i+h} (a_i \times a_j) = 0$; as t runs over the n-th roots of unity, the coefficient of each power t^p modulo $t^n - 1$ vanishes:

$$\sum_{h=0}^{2} c_{p-h,j,h}\,(a_{p-h} \times a_j) = 0 \qquad (j, p \in \mathbb{Z}/n,\ p - h \neq j). \tag{3}$$

The unknown $c_{ijh} = c_{jih}$ appears only in the equations with $i + j + h = \lambda$ fixed, so the system splits by λ. For fixed λ consider the graph on $\mathbb{Z}/n$ with an edge $\{i, \lambda - h - i\}$ for each $h \in \{0, 1, 2\}$ and each i, loops removed; the edge carries $c_{i,\lambda-h-i,h}$, and (3) at the vertex j involves exactly its incident edges, with pairwise independent covectors $a_k \times a_j$. So at a vertex where at most two unknowns are not yet known to vanish, all of them vanish. The reflections $i \mapsto \lambda - h - i$ for $h = 0, 1$ compose to $i \mapsto i + 1$, so the graph is connected, and $2i \equiv \lambda - h$ has a solution for some $h \in \{0, 1\}$, which gives a vertex with a removed loop and at most two edges. Starting there and moving along edges, every vertex has at most two unknowns left, so all c_{ijh} vanish. The relevant minors are therefore nonzero polynomials, and general rational points satisfy all the finitely many open conditions at once. □

We checked the specialization of the proof for $m = 6$ and $3 \le n \le 8$ over $\mathbb{F}_p$ with $p \equiv 1 \pmod{n}$: the radical map has rank d, and rank below d after removing one point. As a check, for $2 \le m \le 30$ the radical map of these integer points has rank d modulo 1000003; the scripts accompany this note.

3. Consequences for sums of squares

Corollary 4. $\mathrm{BSR}(m,3) \ge 2m$ *and* $\mathrm{BSR}(m,4) \ge \lfloor 8m/3 \rfloor$. *For* $m \ge 15$,

$$\mathrm{BSR}(m,4) - z_2(m,4) \;\ge\; \Big\lfloor \frac{8m}{3} \Big\rfloor - \Big\lfloor \frac{5m+6}{2} \Big\rfloor \;\ge\; \frac{m-22}{6}.$$

The last expression is positive for $m \ge 23$, *and the middle one equals* $56 - 55 = 1$ *at* $m = 21$ *and* 0 *at* $m = 22$. *The same bounds hold with* z_{2A} *in place of* z_2.

Proof. The first two bounds are [1, Theorem 1.3] with Theorems 1 and 2; the first was already known as $z_2(m,3) = 2m$ [5]. The difference uses [2, Theorem 1.1]. For z_{2A}, for $m \ge 15$,

$$\Big\lfloor \frac{5m+6}{2} \Big\rfloor = z_2(m,4) \le z_{2A}(m,4) \le \Big\lfloor \frac{4m + z(m,4)}{2} \Big\rfloor = \Big\lfloor \frac{5m+6}{2} \Big\rfloor,$$

by [2, Theorem 1.1], the cell bound [7, Proposition 2.5] and $z(m,4) = m+6$ for $m \ge 6$. □

Corollary 5. *For every fixed* $n \ge 4$, $\mathrm{BSR}(m,n) - z_{2A}(m,n) \ge \frac{n-3}{6}\, m - O_n(1)$, *and the same holds for* z_2.

Proof. By Theorem 3, $\mathrm{BSR}(m,n) \ge mn - n\lceil m/3 \rceil \ge mn - \frac{n(m+2)}{3}$. On the other side, $z_2 \le z_{2A} \le \frac{mn + z(m,n)}{2}$ by the cell bound, and a C_4-free bipartite graph with row degrees d_i has $\sum_i \binom{d_i}{2} \le \binom{n}{2}$, so $z(m,n) = \sum_i d_i \le m + \binom{n}{2}$. Subtracting gives $\frac{n-3}{6} m - \frac{2n}{3} - \frac{1}{2}\binom{n}{2}$. □

So in width four and beyond, the second order and augmented Zarankiewicz numbers miss a linear part of BSR. By the same cell bound no form arising from a simple configuration with a C_4-free one-edge graph reaches $\lfloor 8m/3 \rfloor$ for $m = 21$ or $m \ge 23$: the forms of Corollary 4 are of a different kind.

Theorem 6. *Let* $k = \mathrm{qp}(X)$ *and* $e = mn - k$. *There is a rational biquadratic form* F *with* $\ell_{\mathbb{R}}(F) = e$ *and* $\ell_{\mathbb{Q}}(F) = e + 3$.

Proof. There are k independent rational simple tensors p_i with $D \mapsto (Dp_1, \dots, Dp_k)$ injective on $\mathcal{D}$, since this is a nonempty Zariski open condition and rational points are dense. The face of the forms vanishing at these points is the mechanism of [1, Lemma 3.18]. Let $u_1, \dots, u_e$ be a rational basis of the annihilator of the p_i, read as bilinear forms, and $F = u_1^2 + \cdots + u_{e-1}^2 + 7u_e^2$. A positive semidefinite Gram matrix G of F has $p_i^{\top} G p_i = F(p_i) = 0$, hence $Gp_i = 0$; the difference of two such lies in $\mathcal{D}$ and kills all p_i, so the positive semidefinite Gram matrix is unique, of rank e. Its nondegenerate part is $(e-1)\langle 1 \rangle \perp \langle 7 \rangle$ over $\mathbb{Q}$, so a representation by r rational squares is an embedding of this form in $r\langle 1 \rangle$, and by Witt cancellation it exists exactly when 7 is a sum of $r - e + 1$ rational squares. Since 7 is a sum of four rational squares and not of three, $\ell_{\mathbb{Q}}(F) = e + 3$. □

4. Widths five and six

Let $s_0(m,n)$ be the least s with $g(s) \ge d$ in (1), so $\mathrm{qp} \ge s_0$ always.

Proposition 7. $\mathrm{qp}(\mathbb{P}^{m-1}\times\mathbb{P}^4)=s_0(m,5)$ *for* $5\le m\le 18$ *and* $\mathrm{qp}(\mathbb{P}^{m-1}\times\mathbb{P}^5)=s_0(m,6)$ *for* $6\le m\le 14$*, with the values below.*

Proof. For each size, s_0 integer points give a radical map of rank d modulo 1000003 and modulo 1000033; a nonzero maximal minor modulo a prime is nonzero over $\mathbb{Q}$, so $\mathrm{qp}\le s_0$. The points are in the file `qp_anchura56_cert.json`. □

m, $n=5$	5	6	7	8	9	10	11	12	13	14	15	16	17	18
qp	8	10	12	13	15	16	18	19	21	22	24	25	27	29
BSR $\ge$	17	20	23	27	30	34	37	41	44	48	51	55	58	61

m, $n=6$	6	7	8	9	10	11	12	13	14
qp	12	14	15	17	19	21	23	25	26
BSR $\ge$	24	28	33	37	41	45	49	53	58

In width four the analogue fails: $\mathrm{qp}(\mathbb{P}^6\times\mathbb{P}^3)=10>s_0(7,4)=9$. The values above give $\mathrm{BSR}(m,5)>z_{2A}(m,5)$ for $14\le m\le 18$ and $\mathrm{BSR}(14,6)>z_{2A}(14,6)$, since $mn-\mathrm{qp}$ exceeds the cell bound $\lfloor(mn+m+\binom{n}{2})/2\rfloor$ there.

Remark 8. If $\mathrm{qp}(\mathbb{P}^{m-1}\times\mathbb{P}^{n-1})=s_0(m,n)$ for all m, as these computations suggest for $n=5,6$, then $\mathrm{BSR}(m,n)/m$ is at least $1+\sqrt{(n-1)(n-2)/2}$ asymptotically, that is $1+\sqrt{6}\approx 3.45$ for $n=5$ and $1+\sqrt{10}\approx 4.16$ for $n=6$, against $(n+1)/2$ for the cell bound. For $n=4$ the same expression gives $1+\sqrt{3}$, above the slope $8/3$ that Theorem 2 actually gives.

5. An upper bound

Theorem 9. *For* $m,n\ge 2$, $\mathrm{BSR}(m,n)\le mn-\max(m,n)+1$. *In particular* $2m\le\mathrm{BSR}(m,3)\le 2m+1$ *for* $m\ge 3$ *and* $8\le\mathrm{BSR}(4,3)\le 9$.

Proof. Let $m\ge n$ and $f=\sum_k b_k^2$; let $v_{ij}\in\mathbb{R}^r$ be the coefficient vector of x_iy_j in $(b_1,\dots,b_r)$. The vectors v_{ij} with $j\ge 3$ span a space W of dimension at most $m(n-2)$. After an orthogonal change of the r coordinates, W lies in the first $\dim W$ of them, and the remaining coordinates give bilinear forms in x and y_1,y_2 only. Their sum of squares is a form of size $m\times 2$, hence a sum of $m+1$ squares [1, Theorem 1.4]. So f is a sum of $m(n-2)+m+1$ squares. For $m<n$ exchange the roles of x and y. The lower bound for $n=3$ is Corollary 4. □

This is the bound of [1, Corollary 2.15] for the product of simplices: the mn lattice points are covered by $mn-\max(m,n)$ parallel lines. It improves the general bound $mn-1$ used in [8], where 9 is proved only for a subclass. For $n\ge 5$ it is not the best known: [1, Theorem 2.1] gives $\binom{\mathrm{BSR}(m,n)+1}{2}<\binom{m+1}{2}\binom{n+1}{2}$, which we checked is strictly smaller for every pair $5\le n\le m\le 20$; for $n\le 4$ it never beats Theorem 9.

Questions

Is there a closed formula for $\mathrm{qp}(\mathbb{P}^{m-1}\times\mathbb{P}^{n-1})$ for $n\ge 5$? Is $\mathrm{BSR}(4,3)$ equal to 8 or 9? Han, Jung and Park observe that all known cases satisfy equality in $\mathrm{py}\ge N+1-\mathrm{qp}$ [4, p. 3]; for $\mathbb{P}^{m-1}\times\mathbb{P}^2$ this would give $\mathrm{BSR}(m,3)=2m$. For fixed n, does $\mathrm{BSR}(m,n)/m$ have a limit, and what is it? Asymptotically it lies between $2n/3$ and $\min\{n-1,\sqrt{n(n+1)/2}\}$.

Data and acknowledgement

The scripts that compute the ranks accompany this note. We worked with an AI assistant.

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